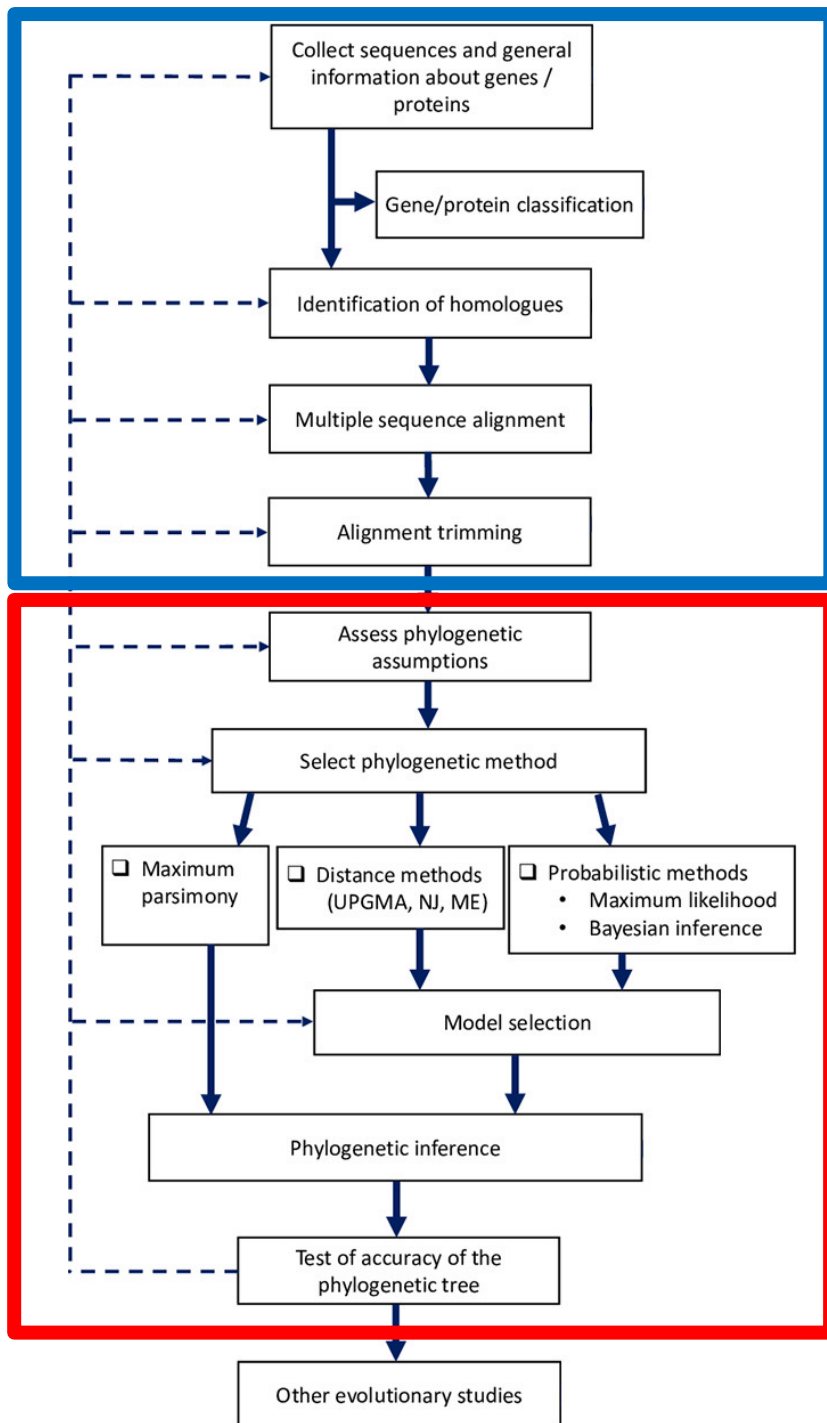

Lecture 1.3

Phylogenetic Data



1. Data preparation

- Taxon and gene sampling
- Sequence alignment (if needed)
- Data filtering

2. Phylogenetic inference

- Model selection
- Estimation of tree
- Further analysis and interpretation

Phylogenetic data

- **Select data to optimise signal:noise**
 - Slowly evolving markers for deep evolutionary events
 - Rapidly evolving markers for recent evolutionary events
- **Minimise violations of phylogenetic assumptions**
 - Some data sets can yield biased phylogenetic estimates
- **Take advantage of existing resources**



Data types

- **Sequence data**
 - Nucleotides
 - Amino acids
- **Binary data** (presence/absence of genomic features)
- **Microsatellites** (repeat numbers)
- **Single-nucleotide polymorphisms (SNPs)**
- **Reduced-representation sequences**

Morphological data

- Morphological characters from extant and extinct taxa

Current Biology

Volume 25, Issue 19, 5 October 2015, Pages R922–R929

Review

Morphological Phylogenetics in the Genomic Age

Michael S.Y. Lee^{1,2},  , Alessandro Palci^{1,2}

Sequence data

- **Coding sequences**
 - Ribosomal RNA
 - Protein-coding genes
- **Non-coding sequences**
 - Intergenic sites
 - Introns
- **Amino acid sequences**

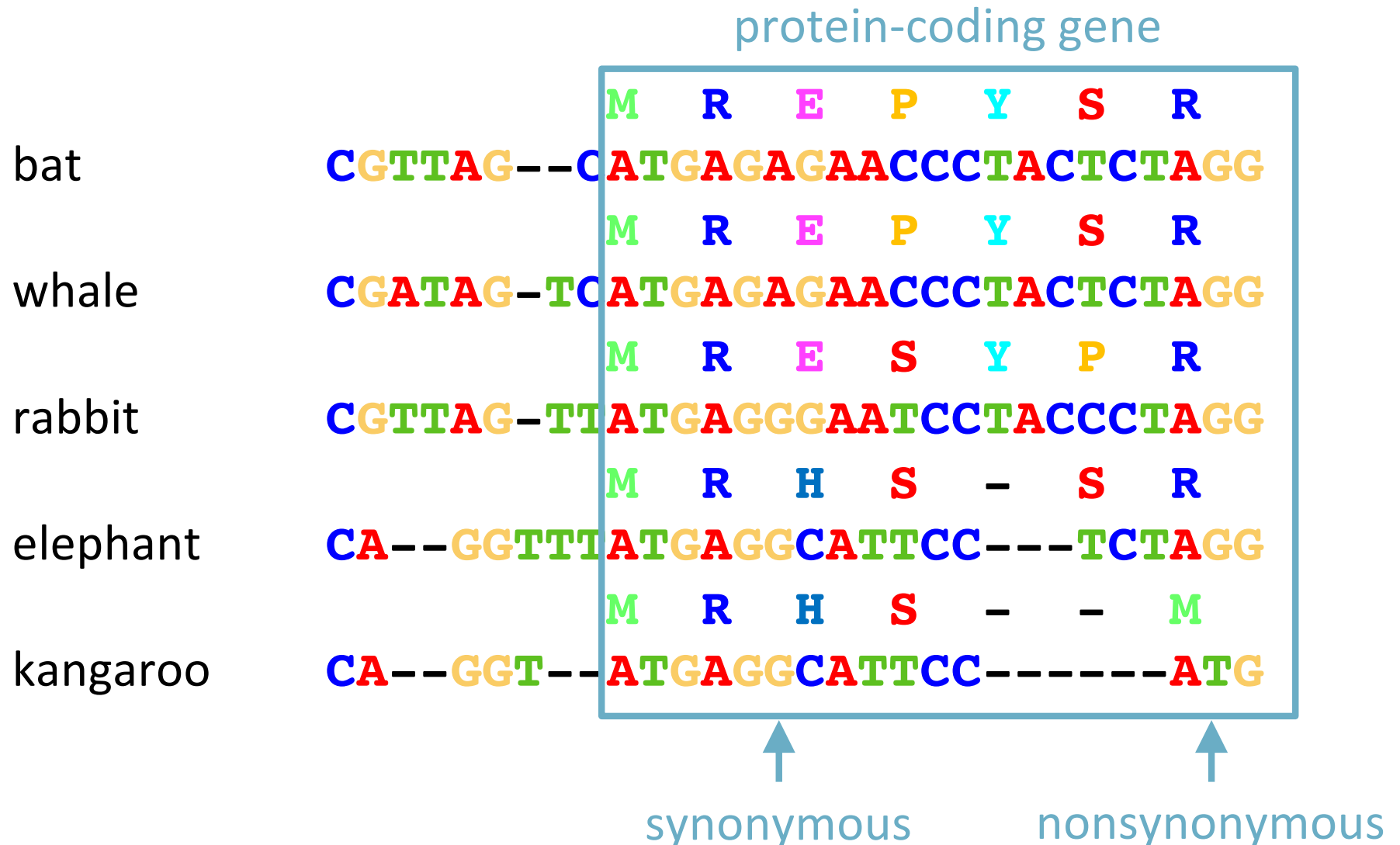


Sequence data

non-coding region

bat	CGTTAG--CATGAGAGAACCCCTACTCTAGG
whale	CGATAG--TCATGAGAGAACCCCTACTCTAGG
rabbit	CGTTAG--TTATGAGGGAATCCTACCCTAGG
elephant	CA--GGTTTATGAGGCATTCC---TCTAGG
kangaroo	CA--GGT---ATGAGGCATTCC-----ATG

Sequence data



Data partitioning

- Sites evolve at different rates
 - Separate substitution model for each gene and codon position?
-

	Gene A	Gene B	Gene C
bat			
whale			
rabbit			
elephant			
kangaroo			

- **Biological**

- Genome
- Genes
- Codon positions
- RNA stems vs loops
- Hydrophobic vs hydrophilic

- **Statistical**

PartitionFinder

- Too many possible partitioning schemes
 - 15 schemes for 4 genes
 - 52 schemes for 5 genes
 - 203 schemes for 6 genes

PartitionFinder 2: New Methods for Selecting Partitioned Models of Evolution for Molecular and Morphological Phylogenetic Analyses □

[Robert Lanfear](#) □, [Paul B. Frandsen](#), [April M. Wright](#), [Tereza Senfeld](#), [Brett Calcott](#)

Molecular Biology and Evolution, Volume 34, Issue 3, March 2017, Pages 772–773,

Gaps and missing data

- **Delete sites with any missing data**
 - Potential loss of informative data
 - Problematic in analyses of data supermatrices
- **Treat gaps as unresolved data**
 - Gap is simultaneously A, C, G, and T
 - Most common approach
- **Code gaps as binary characters**

Gaps and missing data

- Impact of missing data remains poorly understood
- Filter data according to chosen threshold of missing data

	Gene 1	Gene 2	Gene 3	Gene 4	Gene 5
Taxon 1					
Taxon 2					
Taxon 3					
Taxon 4					
Taxon 5					
Taxon 6					

Maximise
gene
sampling

Maximise taxon sampling

Mutational saturation

- Some sites can evolve very rapidly
 - 3rd codon positions
 - Loop regions in RNA
- Multiple hits can erode phylogenetic signal

Excluding Loci With Substitution Saturation Improves Inferences From Phylogenomic Data

David A Duchêne , Niklas Mather, Cara Van Der Wal, Simon Y W Ho

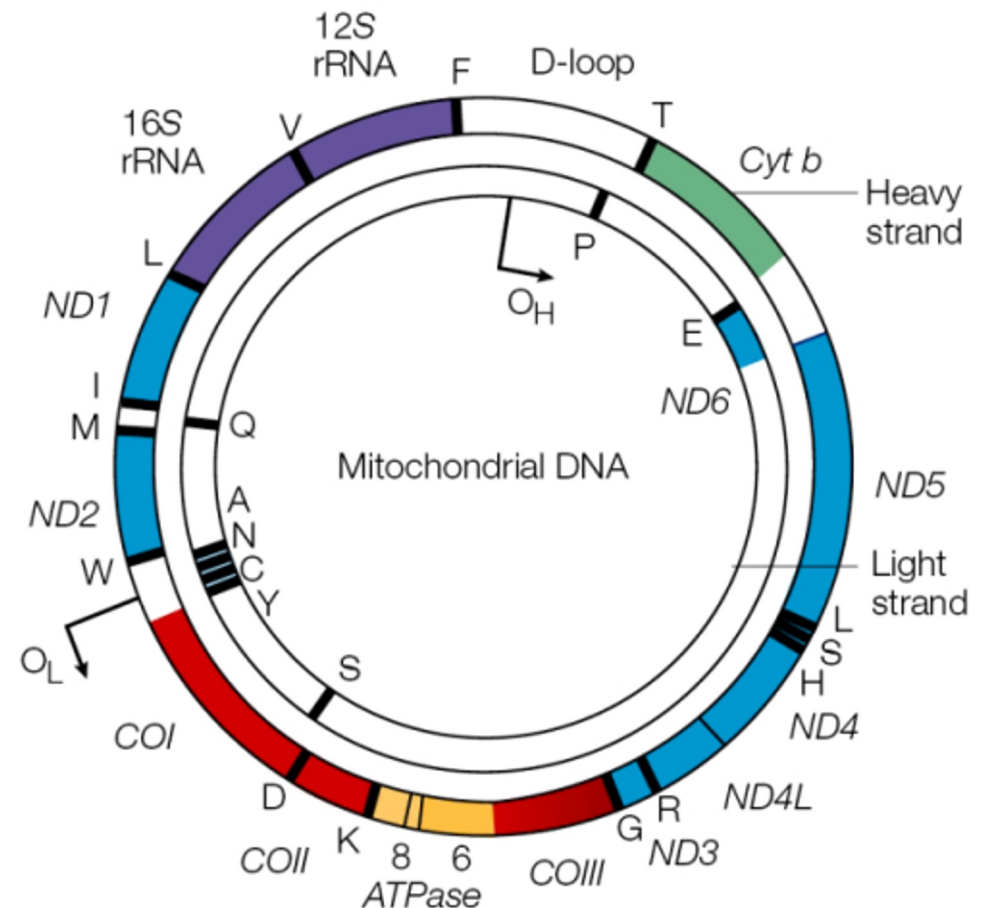
Systematic Biology, Volume 71, Issue 3, May 2022, Pages 676–689,

Saturated sites can be removed to improve signal:noise

High-Throughput Data

Mitochondrial genomes

- Maternally inherited
- Protein-coding genes (e.g., *COI*)
- RNA genes (e.g., *12S*, *16S*)
- Control region



Single-nucleotide polymorphisms

- Single sites sampled from throughout the genome
- More common in intraspecific (population) studies
- Issues to consider:
 - **Recombination**
SNPs are usually unlinked so they are likely to have different (gene) trees
 - **Ascertainment bias**
SNPs are selected for variability and this can mislead estimates of population sizes, rates, and other parameters

Reduced-representation sequences

- Markers identified by cutting genome with restriction enzymes
- Process creates binary data and short sequences
- Examples include RADseq and DArTseq
- Issues to consider:
 - **Recombination**
Markers are usually unlinked so they are likely to have different (gene) trees
 - **Missing data**
Typically a large proportion of missing data



Transcriptomes and exon capture

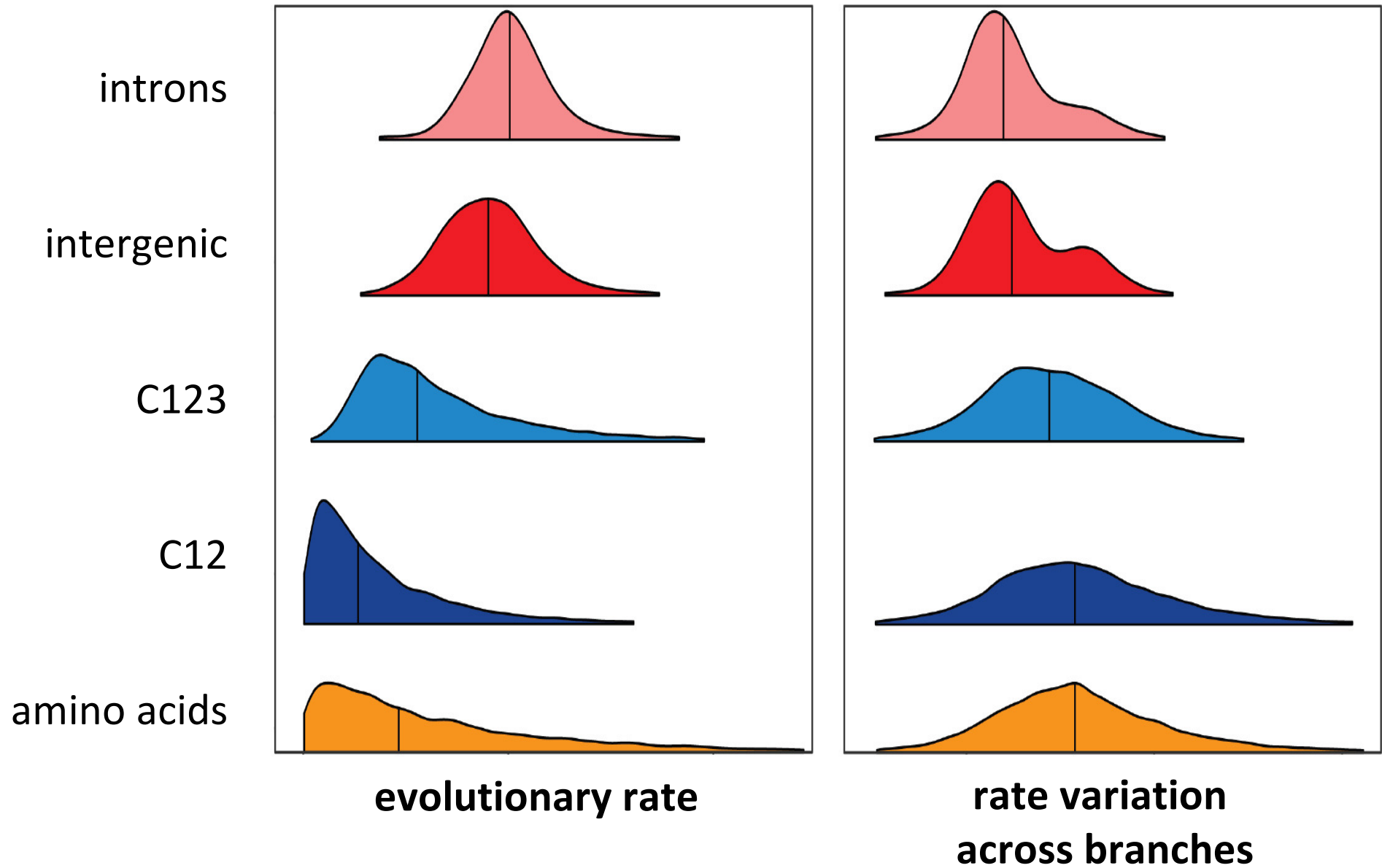
- Large panels of protein-coding loci
- Sequences are easier to align
- Good for inferring deep relationships
- Issues to consider:
 - **Variability**
Might not be much variation at the population level
 - **Selection**
Differences in selection will lead to rate differences across exons

Whole-genome sequencing

- Choice of data types
- Need to reduce computational burden
- **Issues to consider**
 - Single-copy genes
 - Selectively neutral
 - Unlinked loci



Bird genomes



Useful references

